

Grade 7
Unit 4
Lesson 7 Practice

Name _____
Date _____
Period _____

For questions 1 – 10, use 3.14 for π and round to the nearest tenth.

Given the circumference, determine the diameter.

1. 54.95 ft

$$\begin{aligned}C &= \pi \cdot d \\ \frac{C}{\pi} &= d \\ \frac{54.95}{3.14} &= d \\ 17.5 \text{ ft} &= d\end{aligned}$$

2. 195.936 mm

$$\begin{aligned}C &= \pi \cdot d \\ \frac{C}{\pi} &= d \\ \frac{195.936}{3.14} &= d \\ 62.4 \text{ mm} &= d\end{aligned}$$

3. 146.64 in

$$\begin{aligned}C &= \pi \cdot d \\ \frac{C}{\pi} &= d \\ \frac{146.64}{3.14} &= d \\ 46.7 \text{ in} &= d\end{aligned}$$

Given the diameter, determine the circumference.

4. 11 ft

$$\begin{aligned}C &= \pi d \\ C &= 3.14 \cdot 11 \\ C &\approx 34.54 \text{ ft}\end{aligned}$$

5. 7.4 in

$$\begin{aligned}C &= \pi d \\ C &= 3.14 \cdot 7.4 \\ C &\approx 23.236 \text{ in}\end{aligned}$$

6. 29.1 cm

$$\begin{aligned}C &= \pi d \\ C &= 3.14 \cdot 29.1 \\ C &\approx 91.374 \text{ cm}\end{aligned}$$

Given the circumference, determine the radius.

7. $25.12in$

$$\begin{aligned}C &= 2\pi r \\ \frac{C}{2\pi} &= r \\ \frac{25.12}{2 \cdot 3.14} &= r \\ \frac{25.12}{6.28} &= r \\ 4 \text{ in} &\approx r\end{aligned}$$

8. $317.77m$

$$\begin{aligned}C &= 2\pi r \\ \frac{C}{2\pi} &= r \\ \frac{317.77}{2 \cdot 3.14} &= r \\ \frac{317.77}{6.28} &= r \\ 50.6 \text{ m} &\approx r\end{aligned}$$

9. $111.78ft$

$$\begin{aligned}C &= \pi \cdot d \\ \frac{C}{\pi} &= r \\ \frac{111.78}{3.14} &= r \\ \frac{111.78}{3.14} &= r \\ 35.6 \text{ ft} &\approx r\end{aligned}$$

10. A bicycle wheel has a diameter of 26 inches. What is the circumference of the bicycle wheel?

$$\begin{aligned}C &= \pi \cdot d \\ C &= 3.14 \cdot 26 \\ C &= 81.64 \\ C &\approx 81.64 \text{ in}\end{aligned}$$